Fabrication of Oxide/FeSe multilayer films using the PLD technique

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Abstract

In this study, we demonstrate the successful fabrication of TiO_2/FeSe/$SrTiO_3$ and TiO_2/FeSe/$LaAlO_3$ heterostructures using PLD. When the growth rates were sufficiently low, anatase TiO_2, which has been reported to induce superconductivity in FeSe, could be grown epitaxially on FeSe. For TiO_2/FeSe/$SrTiO_3$ with FeSe layers thicker than 5 nm, the FeSe/TiO_2 interface was clean, as confirmed by the observation of Laue fringes around the FeSe(001) reflection. Although a thinner FeSe layer is expected to enhance T_c, we could not observe such behavior because it was difficult to realize interfacial superconductivity while growing anatase TiO_2. Most strikingly, superconductivity appeared when TiO_2 was deposited on FeSe/$LaAlO_3$, clearly demonstrating that the interaction between the top oxide layer and underlying FeSe can induce superconductivity. This work demonstrates the feasibility of fabricating FeSe/oxide superlattices by PLD, establishing a novel platform for the exploration of interfacial superconductivity in iron-based superconductors.

I. INTRODUCTION

Fe-chalcogenide superconductors have attracted much attention because of their high tunability of the superconducting transition temperature (T_c). Although the T_c of bulk FeSe is 9 K at ambient pressure [1], it can be significantly enhanced by compressive strain [2–4], chemical pressure [5], hydrostatic pressure [6, 7], and electron doping through intercalation [8, 9] or the electric-field effect [10–16]. In particular, a zero-resistance temperature (T_c^{zero}) as high as 46 K has been achieved in Refs. [15, 16].

Intriguingly, ultrathin FeSe films grown on typical oxide substrates such as $SrTiO_3$ (STO) [17] and TiO_2 [18, 19] also exhibit enhanced superconductivity owing to interfacial effects. Electron transfer [20–22] and interfacial electron–phonon coupling [23–25] have been reported to play essential roles in this enhancement. For FeSe/STO, angle-resolved photoemission spectroscopy (ARPES) measurements revealed a gap opening at the Fermi level as high as 65 K [26]. However, it remains unclear whether this observation is a direct signature of superconductivity, since transport measurements indicate an onset T_c (T_c^{onset}) around 40 K, which is much lower than the gap-opening temperature [17, 27, 28]. Furthermore, the resistance drop is typically broad, and the zero-resistance state emerges only at T_c^{zero} = 10–30 K, well below T_c^{onset}. This unexpectedly low T_c^{zero} has been attributed to the Berezin-

skii–Kosterlitz–Thouless (BKT) transition[17] or strong inhomogeneity effects[29], both of which originate from the two-dimensional (2D) nature of superconductivity in FeSe/STO. Therefore, suppressing the 2D character while keeping the presence of the interfaces is a promising approach to achieving a higher T_c^{zero} in this system. One possible strategy is to fabricate a superlattice composed of alternating FeSe and oxide layers. In such a structure, if the intermediate FeSe layers are thinner than the coherence length, superconductivity at each FeSe/oxide interface may be Josephson-coupled and reduce the 2D nature. In addition, interfacial electron–phonon coupling may be enhanced, leading to a higher T_c [30]. So far, superlattices based on FeSe have been fabricated using FeSe and FeTe[31], which are chemically and structurally similar compounds. However, superlattices composed of FeSe and oxide materials are more challenging, because direct growth of oxide layers on FeSe is difficult due to its high sensitivity to oxygen. As a first step toward realizing such superlattices, it is important to construct oxide/FeSe/oxide multilayer structures(Fig. 1(a)), in which superconducting layers are confined near each interface. Even in this structure, superconductivity originating at the two interfaces can be coupled to form a Josephson junction, leading to the suppression of 2D nature and an enhancement of T_c if the FeSe interlayer is sufficiently thin.

Pulsed laser deposition (PLD) offers a promising route to deposit oxide materials without oxidizing FeSe. In the PLD growth process, a high-energy laser pulse ablates the target material and generates vaporized species that retain the stoichiometry of the target. As a result, PLD can produce thin films that nearly reproduce the target composition[32]. This feature of PLD allows the growth of epitaxial oxide films under vacuum conditions. Indeed, growth of STO under vacuum has been achieved by tuning laser condition [33]. Furthermore, different materials can be deposited simply by switching the target, which is advantageous for exploring high-T_c superconductivity in FeSe/oxide systems via interface engineering.

In this study, we demonstrate the successful fabrication of TiO_2/FeSe/STO and TiO_2/FeSe/$LaAlO_3$ heterostructures using PLD. When the growth rates were sufficiently low, anatase TiO_2, which has been reported to induce superconductivity in FeSe [18], could be grown epitaxially on FeSe. For TiO_2/FeSe/STO, the FeSe/TiO_2 interface was relatively clean, as confirmed by the observation of Laue fringes in FeSe(001) reflection. With FeSe thicknesses $d_{FeSe} > 5$ nm, no further enhancement of superconductivity was observed, probably because the observed T_c originates from the FeSe/STO interface. Although a thinner FeSe layer is expected to

enhance T_c, we could not observe such behavior because it was difficult to realize interfacial superconductivity while growing anatase TiO_2. Most strikingly, however, superconductivity appeared when TiO_2 was deposited on FeSe/$LaAlO_3$, clearly demonstrating that the interface between the top oxide layer and underlying FeSe can induce superconductivity. This work demonstrates the feasibility of fabricating FeSe/oxide superlattice by PLD, establishing a novel platform for the exploration of interfacial superconductivity in iron-based superconductors.

II. EXPERIMENTAL METHODS

We used STO and $LaAlO_3$(LAO) substrates for the growth of oxide/FeSe/oxide heterostructures. Prior to PLD deposition, surface treatments were performed on both substrates. STO substrates were annealed in air and subsequently soaked in water, following the detailed procedure described in previous reports [34]. For LAO, the substrates were first annealed at 1000 °C for 12 hours in air, followed by quenching from 500 °C to room temperature (~ 27°C) during cooling to obtain an atomically flat surface. After quenching, the substrates were soaked in deionized water for two days to achieve AlO_2^- termination [35]. FeSe films were grown on STO and $LaAlO_3$ substrates at a substrate temperature of 500 °C with a laser repetition rate of 1 Hz under high vacuum ($\sim 1.0 \times 10^{-8}$ Torr). As the top oxide material, we employed TiO_2. The anatase phase of TiO_2 has an in-plane lattice constant comparable to that of FeSe ($a = 3.78$ Å for TiO_2 and 3.81 Å for FeSe/STO), and it has been reported that FeSe on anatase TiO_2 exhibits superconductivity similar to that observed for FeSe/STO [18]. After the growth of FeSe, TiO_2 layers were subsequently deposited at the same substrate temperature under high vacuum ($\sim 1.0 \times 10^{-8}$ Torr), using a pellet fabricated from rutile TiO_2 powder. All films were finally capped with undoped Si to prevent degradation due to air exposure. We note that this Si capping layer does not change the carrier density of TiO_2 or FeSe because it is an intrinsic semiconductor with a very low carrier density. The orientation and crystal structure of the grown films were characterized by X-ray diffraction (XRD) using Cu Kα radiation at room temperature. Film thicknesses were estimated from the Laue fringes around the FeSe(001) reflections or from the deposition time. Details are described in the Supplemental Material[36]. Resistivity measurements were carried out using a Physical Property Measurement System (PPMS) in

the temperature range of 2–300 K.

III. RESULTS

The growth of anatase TiO_2 on FeSe requires careful optimization of deposition parameters. Previous studies of PLD growth of anatase TiO_2 on LAO showed that the deposition rate influences the density of oxygen vacancies [37], which may affect the growth of anatase phase on FeSe. We fabricated TiO_2/FeSe/STO heterostructures while varying the deposition rates of both FeSe (r_{FeSe}) and TiO_2 ($r_{\mathrm{TiO_2}}$). The phase of the top TiO_2 layer was examined using Reflection High-Energy Electron Diffraction (RHEED) just after the deposition (Fig. 1(b)). Because the in-plane lattice constants of FeSe and anatase TiO_2 are nearly identical, the streak spacing in the RHEED patterns is expected to be equivalent after FeSe deposition and after deposition of the top TiO_2 layer. When TiO_2 was deposited at a rate exceeding 0.04 nm/min, the streak spacing clearly deviated from that of FeSe, irrespective of the FeSe growth rate, indicating that the top oxide layer was not anatase TiO_2. Furthermore, we found that when FeSe itself was grown at a rate higher than 0.25 nm/min, the subsequent TiO_2 layer failed to crystallize in the anatase phase, even when the TiO_2 deposition rate was kept below 0.04 nm/min. These observations suggest that both the growth rates of FeSe and TiO_2 must be carefully controlled to suppress competing phases.

Successful growth of anatase TiO_2 was achieved only under the combined conditions of $r_{\mathrm{FeSe}} < 0.25$ nm/min and $r_{\mathrm{TiO_2}} < 0.04$ nm/min, as illustrated in the right panel in Fig. 1(b). The corresponding RHEED image exhibits streaks with identical spacing for FeSe and TiO_2, confirming that the grown oxide phase was anatase. The appropriateness of this low $r_{\mathrm{TiO_2}}$ was confirmed by growing a TiO_2 film on an LAO substrate under similar conditions, where the anatase phase was confirmed by RHEED and XRD measurements (see the Supplemental Material[36]). A systematic summary of the RHEED images obtained at different growth rates is also provided in Fig. 1(c), clearly indicating that both $r_{\mathrm{TiO_2}}$ and r_{FeSe} are important to obtain the anatase phase on FeSe. Figure 1(d) shows the XRD pattern of a representative anatase TiO_2/FeSe/STO heterostructure. Distinct FeSe(00l) diffraction peaks were observed even after deposition of the TiO_2, demonstrating that the crystalline order of FeSe was preserved during the process. In addition, Laue fringes were clearly visible around the FeSe(001) reflection (Fig. 1(e)), indicating flat interfaces between FeSe and TiO_2.

These results demonstrate that careful control of the growth rates ($r_{\mathrm{TiO_2}}$ and r_{FeSe}) enables the fabrication of TiO_2/FeSe/STO heterostructures, while preserving the single crystallinity of FeSe and ensuring flat top oxide/superconductor interfaces.

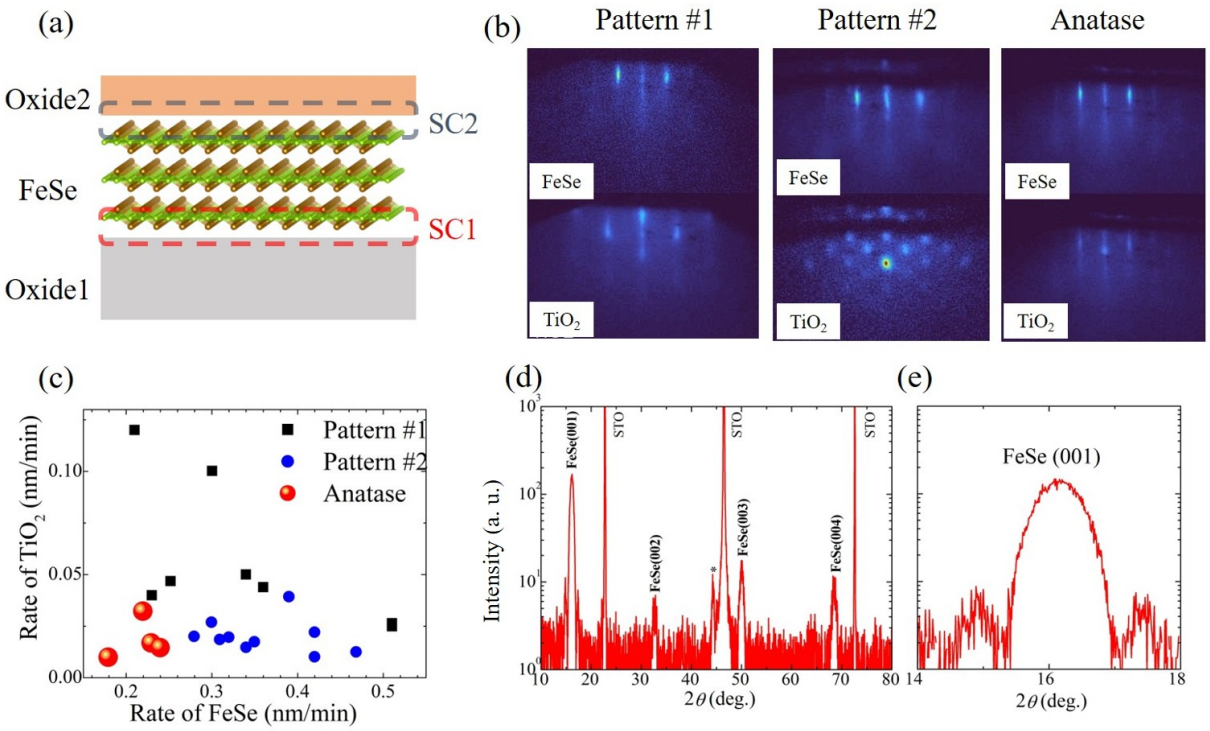


FIG. 1. (a) Schematic illustration of the oxide/FeSe/oxide heterostructure. Superconductivity originating from each FeSe/oxide interface emerges. (b) RHEED images obtained after the growth of FeSe (top) and TiO_2 (bottom). Left, middle, and right images correspond to $r_{\mathrm{TiO_2}} > 0.04$ nm/min, $r_{\mathrm{TiO_2}} < 0.04$ nm/min with $r_{\mathrm{FeSe}} > 0.25$ nm/min, and $r_{\mathrm{TiO_2}} < 0.04$ nm/min with $r_{\mathrm{FeSe}} < 0.25$ nm/min, respectively. (c) Summary of RHEED patterns for various $r_{\mathrm{TiO_2}}$ and r_{FeSe}. (d) XRD pattern of a representative TiO_2/FeSe/STO heterostructure with $d_{\mathrm{FeSe}} = 10$ nm and $d_{\mathrm{TiO_2}} \sim 2$ nm. The peak indicated by an asterisk is the Fe(101) peak originating from the sample stage. (e) Enlarged plot of the FeSe(001) reflection.

Figure 2(a) shows the temperature dependence of the sheet resistance of TiO_2/FeSe/STO heterostructures and FeSe/STO films with FeSe thicknesses (d_{FeSe}) ranging from 10 to 1.5 nm, while keeping the TiO_2 thickness ($d_{\mathrm{TiO_2}}$) fixed at 2 nm. When $d_{\mathrm{FeSe}} = 10$ and 5 nm, the sheet resistance and residual resistance ratio RRR($= R_{(300\mathrm{K})}/R_{(40\mathrm{K})}$) were almost the same for the TiO_2/FeSe and FeSe/STO films. This suggests that TiO_2 deposition did not cause

significant degradation of FeSe, which is consistent with the presence of Laue fringes around the FeSe(001) reflection. Furthermore, T_c^{onset} was unchanged regardless of the presence of the TiO_2 overlayer (Fig. 2(b)). This result is not surprising, since possible superconductivity at the TiO_2/FeSe and FeSe/STO interfaces is not expected to couple owing to the short coherence length [38]. This indicates that the observed superconductivity originates from the FeSe/STO interface. For thinner FeSe layers, superconductivity in FeSe/STO was slightly suppressed as the FeSe thickness decreased, and the 1.5-nm-thick film did not exhibit superconductivity. This suppression of superconductivity is in contrast to the literature[22, 39] and our previous study[34, 38], which reported an enhancement of superconductivity with decreasing thickness. This discrepancy may be due to the FeSe deposition rate being as low as 0.1 nm/min in the present study, whereas it was 0.2–0.5 nm/min in the previous study[34]. Although such a low deposition rate was necessary for the subsequent growth of anatase TiO_2, it was not optimal for obtaining high-quality ultrathin FeSe films. When comparing the TiO_2/FeSe ($d_{FeSe} = 2.5$ nm) and FeSe ($d_{FeSe} = 3$ nm) films, the sheet resistance increased by a factor of 3.4 and the transition became broader after TiO_2 deposition. At $d_{FeSe} \sim 1.5$ nm, TiO_2/FeSe/STO exhibited higher resistance than FeSe/STO. The resistance of these films exceeded the quantum critical value ($h/(2e)^2$) associated with the superconductor–insulator transition[40], and no superconducting transition was detected. These differences between films with and without a TiO_2 layer may be caused either by a slight difference in thickness or by the deposition of TiO_2 on FeSe. In the latter case, one possible mechanism is structural damage to, or oxidation of, FeSe caused by high-energy species in the laser ablation plume.

To overcome this issue, we employed a different substrate material, LAO, in an attempt to prepare interfaces with reduced degradation. LAO has a lattice constant of $a = 3.78$ Å, which is almost identical to that of anatase TiO_2. Therefore, using an LAO substrate is expected to provide a better interface, as the in-plane lattice constants of anatase TiO_2 and FeSe are more closely matched. Figures 3(a) and (c) show the XRD patterns of FeSe/LAO ($d_{FeSe} = 5.5$ nm)and TiO_2/FeSe/LAO ($d_{FeSe} = 5.5$ nm, $d_{TiO_2} \sim 2$ nm) .

All FeSe (00l) peaks were observed for FeSe/LAO; however, the FeSe(002) and (004) reflections were not observed for TiO_2/FeSe/LAO because of their reduced intensities. Furthermore, Laue fringes were not observed for the film with a TiO_2 overlayer (Fig. 3(d)), in contrast to FeSe/LAO (Fig. 3(b)). These results indicate that the TiO_2/FeSe interface

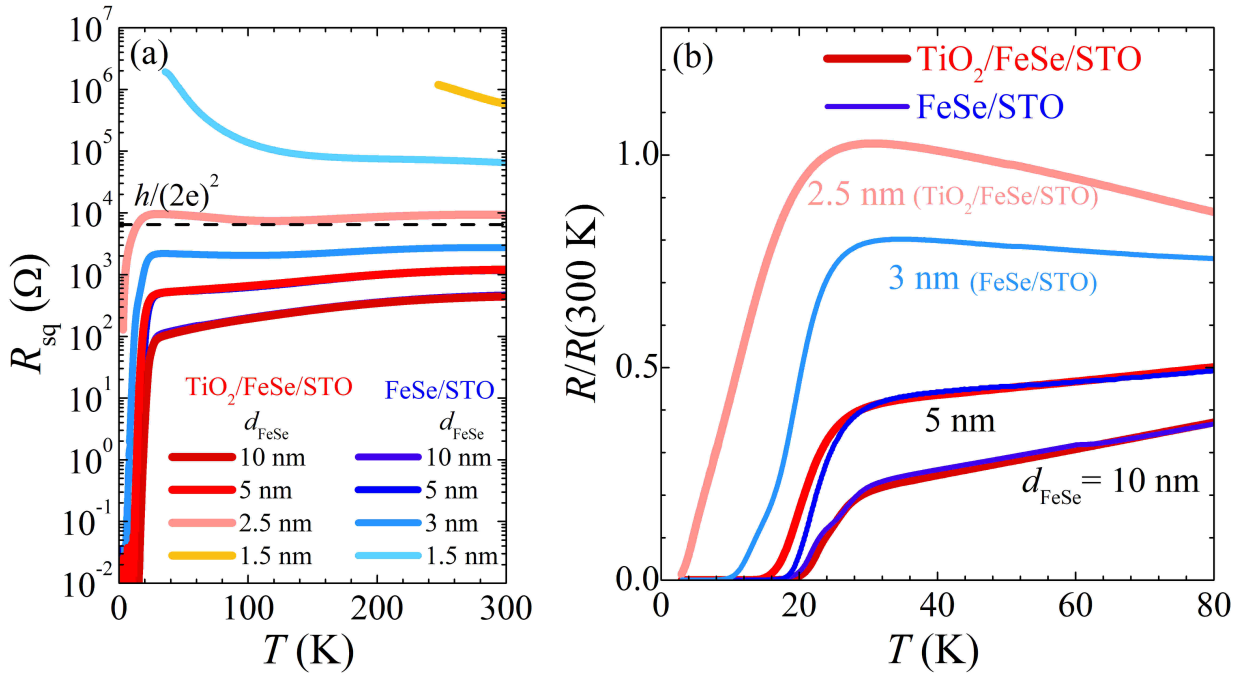


FIG. 2. (a) Temperature dependence of the sheet resistance of TiO_2/FeSe/STO and FeSe/STO films with different d_{FeSe}. For TiO_2/FeSe/STO heterostructures, the TiO_2 thickness was kept at $d_{TiO_2} \sim 2$ nm. (b) Temperature dependence of the normalized resistance of TiO_2/FeSe/STO and FeSe/STO films in the range 0–80 K.

became unexpectedly rough and that deposition of TiO_2 probably introduces disorder into FeSe, as in the case of FeSe/STO. However, in contrast to the films on STO, emergence of superconductivity only after depositing TiO_2 was observed for the films on LAO. Figure 3(e) shows the temperature dependence of the resistance of FeSe/LAO films with and without a TiO_2 overlayer. FeSe/LAO without TiO_2 did not show a superconducting transition down to 2 K. This absence of superconductivity, unlike in FeSe/STO, can be attributed to the lack of electron doping from the substrate. First, the work function of AlO_2^--terminated LAO is calculated to be 6.87 eV[41], which is higher than that of FeSe (5.1 eV), in contrast to the situation in FeSe/STO[21]. Second, although charge transfer driven by polar discontinuity, as in LAO/STO, may in principle occur in LAO-based structures, the B-terminated surface employed in this study would result in hole doping rather than electron doping. In contrast to FeSe/LAO, the film with a TiO_2 overlayer exhibited a superconducting transition at low temperatures. The superconducting origin of the resistive drop was further confirmed by its suppression under applied magnetic fields (Fig. 3(f)). The onset

transition temperature was T_c^{onset} = 16 K, which is higher than that of bulk FeSe. Zero resistivity was observed at 3 K, which may be related to structural damage, as suggested by the absence of Laue fringes around the FeSe(001) reflection. This result suggests that the emergence of superconductivity can be attributed to interfacial effects at the TiO_2/FeSe interface. The detailed mechanism underlying the emergence of superconductivity is still unclear. However, we believe that electron transfer from the oxide to FeSe is likely to be relevant, similar to the case of FeSe/STO. This possibility could be investigated by electron energy-loss spectroscopy (EELS) combined with scanning transmission electron microscopy (STEM)[22]. The different behavior of the films on STO and LAO upon TiO_2 deposition is likely related to the presence of superconductivity originating from the FeSe/substrate interface. In the case of FeSe/STO, superconductivity with $T_c^{onset} \sim 25$ K emerges from the FeSe/STO interface and may mask a possible superconducting transition originating from the TiO_2/FeSe interface with a lower T_c, whereas FeSe/LAO does not exhibit such substrate-induced superconductivity.

The emergence of superconductivity in TiO_2/FeSe/LAO demonstrates that a top oxide overlayer can induce the emergence of superconductivity, analogous to the substrate-induced effect[18]. This observation indicates that coupling between the superconductivity originating at each interface in the TiO_2/FeSe/oxide heterostructure is, in principle, possible. Therefore, our results suggest that constructing a TiO_2/FeSe superlattice provides a promising platform for enhancing or engineering interfacial superconductivity. At this stage, however, the existence of superconductivity originating at the TiO_2/FeSe interface on the STO substrate was not experimentally confirmed because the T_c associated with FeSe/STO may be higher than that associated with the TiO_2/FeSe interface. To reveal the existence of superconductivity near the TiO_2/FeSe interface, AC diamagnetic measurements would be promising. However, because the superconductivity is expected to be inhomogeneous, improving the interface quality is also necessary. In addition, degradation of FeSe during TiO_2 deposition may hinder coupling between the superconducting regions at the two interfaces. In this case, one possible degradation mechanism is damage to FeSe caused by high-energy species in the laser ablation plume. Thus, a promising way to mitigate damage is to reduce the kinetic energy of the plume. This could be achieved by introducing a He buffer gas during PLD[42], which reduces the kinetic energy through scattering between plume species and He atoms, and will be explored in future work.

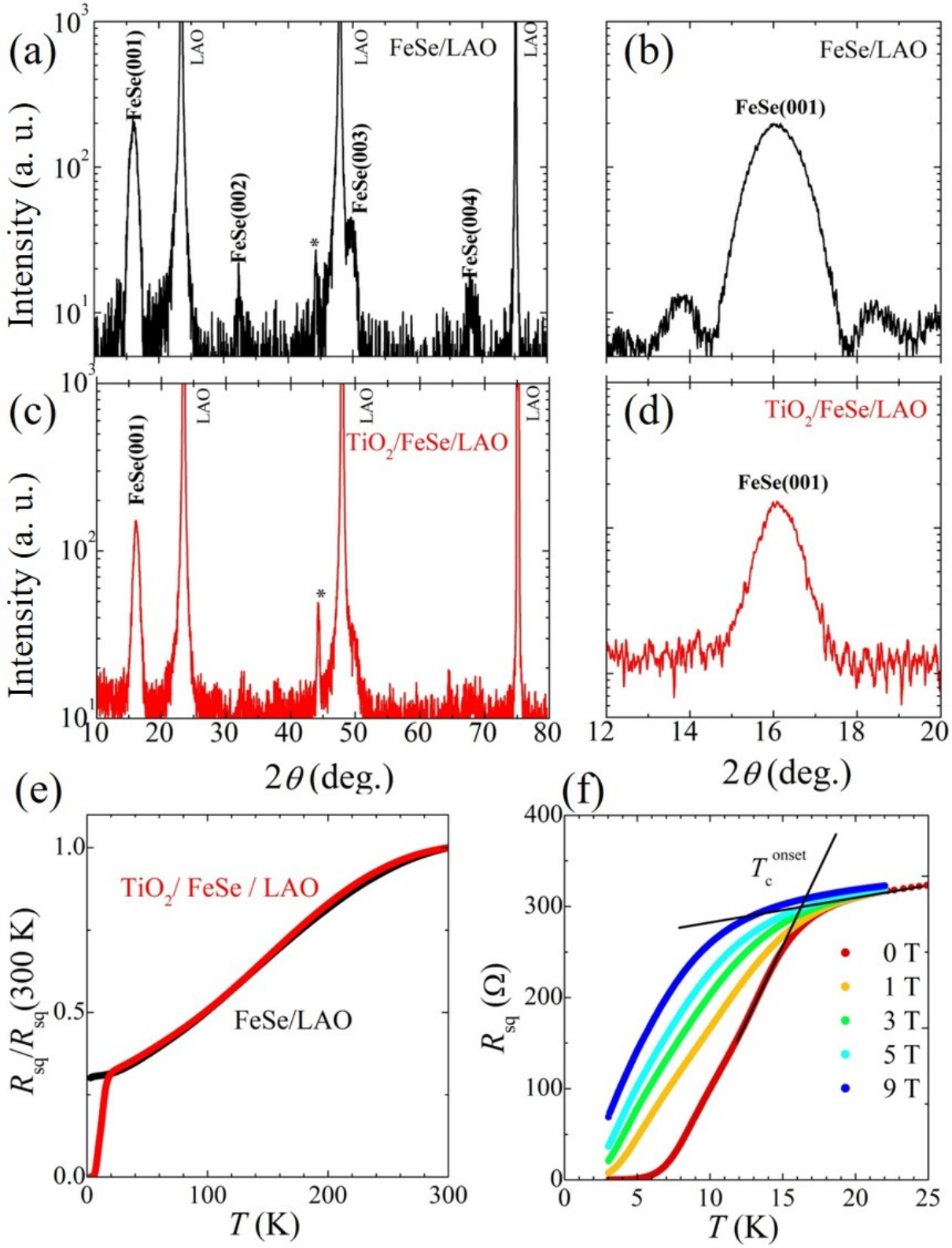


FIG. 3. (a) and (c) XRD patterns of FeSe/LAO and TiO_2/FeSe/LAO, respectively. (b) and (d) show enlarged plots around the FeSe(001) reflections. (e) Temperature dependence of the normalized resistance of TiO_2/FeSe/LAO and FeSe/LAO films with $d_{FeSe} \sim 5$ nm. (f) Temperature dependence of the sheet resistance of TiO_2/FeSe/LAO films under magnetic fields of 0–9 T.

IV. CONCLUSION

In this study, we demonstrate the successful fabrication of TiO_2/FeSe/STO and TiO_2/FeSe/LAO heterostructures using PLD. When the growth rates were sufficiently low, anatase TiO_2, which has been reported to induce superconductivity in FeSe, could be grown epitaxially on FeSe. For TiO_2/FeSe/$SrTiO_3$ with FeSe layers thicker than 5 nm, the FeSe/TiO_2 interface was clean, as confirmed by the observation of Laue fringes around the FeSe(001) reflection. Although a thinner FeSe layer is expected to enhance T_c, we could not observe such behavior because it was difficult to realize interfacial superconductivity while growing anatase TiO_2. Most strikingly, however, superconductivity appeared when TiO_2 was deposited on FeSe/LAO, clearly demonstrating that the interface between the top oxide layer and underlying FeSe can induce superconductivity. This work demonstrates the feasibility of fabricating FeSe/oxide superlattices by PLD, establishing a novel platform for the exploration of interfacial superconductivity in iron-based superconductors.

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